QQ-A-250/3F
September 21, 1978
SUPERSEDING
QQ-A-250/3E
January 18, 1971

FEDERAL SPECIFICATION

ALUMINUM ALLOY ALCLAD 2014, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy alclad 2014 plate and sheet; the general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (2014) core with thin layers of an aluminum alloy (6003) anodic to the core and of approximately equal thickness bonded to both surfaces.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T3, T4, T6, T42, T62, T451, T651, or F temper, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

FSC 9535

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Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Philadelphia, Atlanta, Chicago, Kansas City, MO, Houston, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the alclad plates and sheets shall conform to the requirements specified in table I for core and cladding, respectively.

TABLE I. Chemical composition 1/

Element	Analysis			
	Core Alloy (2014)		Cladding alloy (6003)	
	Minimum Percent	Maximum Percent	Minimum Percent	Maximum Percent
Copper	3.9	5.0	-	0.10
Silicon	0.50	1.2	0.35	1.0
Manganese	0.40	1.2	-	0.8
Magnesium	0.20	0.8	0.8	1.5
Iron	-	0.7	-	0.6
Chromium	-	0.10	-	0.35
Zinc	-	0.25	-	0.20
Titanium	-	0.15	-	0.10
Other elements, each	-	0.05	-	0.05
Other elements, total	-	0.15	-	0.15
Aluminum	Remainder		Remainder	

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1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 in. or 4 times D1/ 2/, minimum
			Minimum	Extension under load	
	Inches	psi	psi	Inch/Inch	Percent
0	0.020 thru 0.499	30,000 3/	14,000 3/	0.0034	16
	0.500 thru 1.000 4/	32,000 3/	---	---	10
T3 5/	0.020 thru 0.039	55,000	34,000	0.0052	14
	0.040 thru 0.249	57,000	35,000	0.0053	15
T4 6/	0.020 thru 0.039	55,000	32,000	0.0051	14
	0.040 thru 0.249	57,000	34,000	0.0052	15
T451	0.250 thru 0.499	57,000	36,000	0.0054	15
	0.500 thru 1.000 4/	58,000	36,000	0.0054	14
	1.001 thru 2.000 4/	58,000	36,000	0.0054	12
	2.001 thru 3.000 4/	57,000	36,000	0.0054	8
T42 7/	0.020 thru 0.039	55,000	32,000	0.0051	14
	0.040 thru 0.099	57,000	34,000	0.0052	15
	0.100 thru 0.499	57,000	34,000	0.0054	15
	0.500 thru 1.000 4/	58,000	34,000	0.0054	14

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TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 in. or 4 times D1/ 2/, minimum
			Minimum	Extension under load	
	Inches	psi	psi	Inch/Inch	Percent
T6 and T62 7/	0.020 thru 0.039	61,000	53,000	0.0073	7
	0.040 thru 0.249	64,000	56,000	0.0075	8
T651 and T62 7/	0.250 thru 0.499	64,000	57,000	0.0075	8
	0.500 thru 1.0004/	67,000	59,000	0.0076	6
	1.001 thru 2.0004/	67,000	59,000	0.0076	4
	2.001 thru 2.5004/	65,000	58,000	0.0076	2
	2.501 thru 3.0004/	63,000	57,000	0.0075	2
	3.001 thru 4.0004/	59,000	55,000	0.0073	1
F	All	8/	8/	8/	8/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ The properties for these thicknesses are those of the core alloy since the tests are made on a round specimen machined from the plate.
5/ Applicable to flat sheet only.
6/ Applicable to coiled sheet only.
7/ Material in the T42 and T62 tempers is not available from the materials producers.
8/ No requirements.

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (0) and the as-fabricated (F) tempers without the subsequent imposition of cold work or forming operations shall, after proper solution heat treatment, also conform to the requirements of table II for the T42 temper. Material as received in the T3, T4, T451, T6, and T651 tempers shall, after proper resolution heat-treatment and natural aging for 4 days at room temperature, be capable of conforming to the requirements specified in table II for the T42 temper. Material in the T3 or T4 tempers shall, after proper aging, be capable of conforming to the requirements specified in table II for the T6 temper. Material in the T42 temper shall, after proper aging, be capable of conforming to the requirements specified in table II for the T62 temper. Material in the T451 temper shall, after proper aging, be capable of conforming to the requirements specified in table II for the T651 temper.

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3.2.3 Bend test. Bend test specimens taken from material shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

TABLE III. Bend test factor "N"

Thickness of material	Tempers		
	0	T3, T4 and T42	T6 and T62
Inch			
0.020 thru 0.039	0	3	4
0.040 thru 0.050	0	3	5
0.051 thru 0.124	0	3	6
0.125 thru 0.249	1	4	8
0.250 thru 0.499	2	5	10

3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy plates that are bonded to the two sides of the aluminum alloy (2014) ingot or slab to form a composite that is to be rolled, shall each have a thickness as specified in table IV. If question arises concerning the thickness of cladding of the finished sheet or plate, samples examined in accordance with QQ-A-250 shall show an average thickness of cladding on each side, not less than that specified in table IV.

TABLE IV. Cladding thickness

Thickness of finished plate or sheet	Nominal cladding thickness per side; percent of composite thickness	Average thickness per side of cladding on finished plate or sheet, minimum percent of plate or sheet thickness
Inches		
Up thru 0.024	10	8
0.025 thru 0.039	7-1/2	6
0.040 thru 0.099	5	4
0.100 thru 4.000	2-1/2	2

3.4 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

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TABLE V. Ultrasonic discontinuity acceptance limits 1/

Thickness	Maximum weight per piece	Discontinuity class 2/
Inches	Pounds	
0.500 thru 1.499	2,000	B
1.500 thru 3.000	2,000	A
3.001 thru 4.000	2,000	B

1/ Discontinuities in excess of those listed in table V may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.

2/ The discontinuity class limits are defined in MIL-I-8950.

3.5 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250.)

4.1 Heat-treatment.

4.1.1 Aging period before testing. Specimens in the T3, T4, T42, and T451 tempers will not be required to be tested within 4 days after completion of the solution heat-treatment. If the manufacturer so elects, samples may be tested after less than 4 days aging; but if they fail to show the specified properties, the test samples shall be discarded and additional samples shall be tested after 4 days aging.

4.2 Mechanical test after heat-treatment.

4.2.1 Number of test after heat-treatment. From material in the annealed (O) and as fabricated (F) tempers, an additional number of specimens equal to those required by QQ-A-250 shall be taken and tested after solution heat-treatment to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY (See QQ-A-250.)

6. NOTES

6.1 Intended use. Alclad 2014 plate and sheet are used where high strength with moderately good resistance to corrosion are required. Alclad 2014 in the T6 or T62 temper has better formability than alclad 7075-T6, but the strength is not as high.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).
(c) Dimensions required.
(d) Requirements for sizes not specifically covered (see QQ-A-250).
(e) Whether ultrasonic inspection of plate is required (see 3.4).
(f) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 To secure maximum resistance to corrosion in the T4 or T451 temper, the material should be quenched in cold water; but for material that is to be artificially aged to the T6 temper, a milder quench is desirable. For that reason, heavy sections have greater inherent resistance to corrosion in the T651 temper because of the slower quench that results from the greater mass of metal. Thinner sheet has a thicker cladding to compensate for the lower resistance to corrosion of the core alloy after artificial aging to the T6 temper.

6.4 Temper. Tempers T451 and T651 are available in plate only. Temper T3 is available in flat sheet only.

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